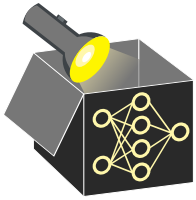
# Interpretable Machine Learning



## Instance-wise Feature Selection

**The movie experience was awful**

↓ Explain

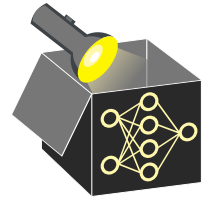
The movie experience **was awful**

↓ Predict

### Learning goals

- Instance-wise feature selection
- Explain then predict models
- Optimizing using explanation data

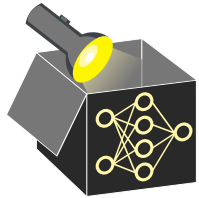
# INSTANCE-WISE FEATURE SELECTION



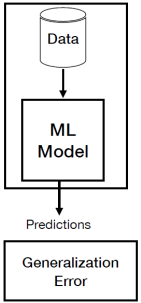
- Select a subset of features conditioned or based on the input instance
  - Two instances might not have the same feature mask

- Instance-wise feature selection similar to feature attribution — important features are selected
- Unambiguous with respect to explanation (more on that later)

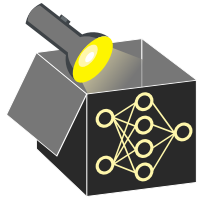
# INSTANCE-WISE FEATURE SELECTION - EXAMPLE IN LANGUAGE



Standard ML



# INSTANCE-WISE FEATURE SELECTION - EXAMPLE IN LANGUAGE



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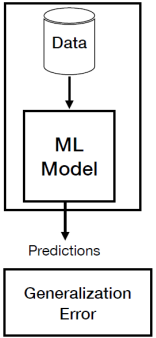
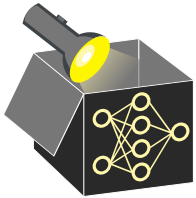


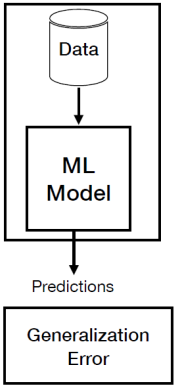

**The movie experience was awful**

Predict

# INSTANCE-WISE FEATURE SELECTION - EXAMPLE IN LANGUAGE



Standard ML

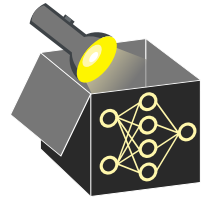



**The movie experience was awful**

Predict

Parameterised Model

# INSTANCE-WISE FEATURE SELECTION - EXAMPLE IN LANGUAGE



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↓ Explain

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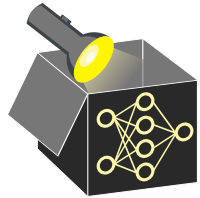

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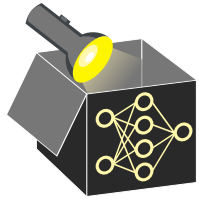

**The movie experience was awful**

Explain

Parameterised Model

The movie experience **was awful**

Predict

Parameterised Model

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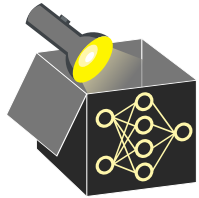


**The movie experience was awful**

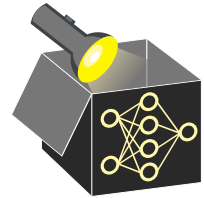
↓ Explain — Parameterised Model

The movie experience **was awful**

Ensure prediction is solely on the explanations

↓ Predict — Parameterised Model

# INSTANCE-WISE FEATURE SELECTION - EXAMPLE IN LANGUAGE



**The movie experience was awful**

Explain

Parameterised Model

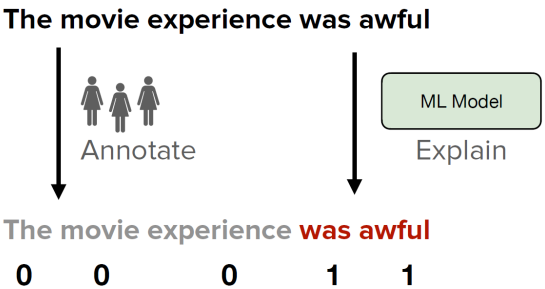
explanation **The movie experience was awful**

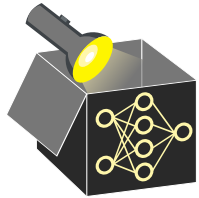
Ensure prediction is solely on the explanations

Predict

Parameterised Model

# EXPLANATION DATA



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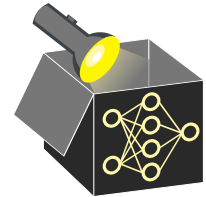

**The movie experience was awful**

Annotate

ML Model

Explain

The movie experience **was awful**

**0 0 0 1 1**

$$\mathcal{L}_{\text{exp}} = \frac{1}{|S|} \sum_{i=1}^{|S|} |S_{t^i}| \cdot \text{BCE}\left(p^i, t^i\right)$$

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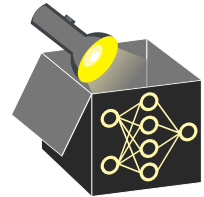


**The movie experience was awful**

ML Model

Explain

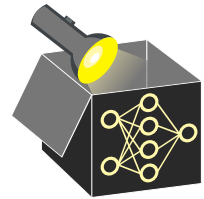
The movie experience **was awful**

**0 0 0 1 1**

$$\mathcal{L}_{\text{exp}} = \frac{1}{|S|} \sum_{i=1}^{|S|} |S_{t^i}| \cdot \text{BCE}\left(p^i, t^i\right)$$

Predictor Model

# EXPLAIN THEN PREDICT USING EXPLANATION DATA



- Selecting features conditioned on individual instances results in better task performance in comparison to global feature selection
- Instance-wise feature selection has higher inherent sparsity
- The output of the feature-selection stage is the explanation
- The predictor depends solely on the masked input and therefore unambiguous with respect to explanation